Pipeline (Set)	Set 1: MTHL Alignment	Set 2: AeroGis Airport	Set 3: Gorakhpur Flood	Set 4: Water Management
Problem	1D Path-Finding: Find the single optimal path (a line) for a bridge.	2D Area Design: Find the best site (a rectangle) and then the optimal layout within it.	3D Urban Intervention: Find multiple interventions (ponds, levees) for a high-risk urban zone.	3D Water Resource: A refactor of Set 3. Finds multiple high-risk zones, then designs interventions for each.
Orchestrator	run_alignment_analysis.py (3-Stage)	run_aero_design.py (3-Stage)	run_gorakhpur_analysis.py (3-Stage)	run_water_management_analysis.py (4-Stage)
Stage 1: Diagnose	alignment_diagnostics.py (GIS Cost Atlas)	airport_selector.py (GIS Area Search & Siting)	urban_diagnostics.py (GIS Cost Atlas)	water_resource_diagnostics.py (GIS Cost Atlas)
Stage 2: Focus	(N/A - Path is predefined A to B)	(N/A - Siting is part of S1)	priority_zone_extractor.py (Identifies one high-risk zone)	water_opportunity_extractor.py (Identifies multiple zones)
Stage 3: Plan (AI)	alignment_planner.py (GA for path coordinates)	genetic_designer.py (GA for layout components)	intervention_planner.py (GA for multiple interventions)	water_intervention_planner.py (GA for multiple interventions)
Stage 4: Validate (Physics)	(validator script mentioned)	bioswale_designer.py (Bayesian Opt. + hrf.py solver)	detailed_engineer.py (Bayesian Opt. + hrf.py solver)	detailed_water_engineer.py (Bayesian Opt. + hrf.py solver)
Key Output	.kml / .shp of the best bridge path.	.glb 3D CAD model of a full airport with flood defenses.	.glb 3D CAD model of a city zone with flood defenses.	.glb 3D CAD model of multiple optimized water interventions.